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Kyoei: Toward Repairable Collaboration Across Organizations for People and Their Agents

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Abstract. Organizations and their agents build on work that another organization can later withdraw or correct. Git records what changed and who changed it. It does not record which work relies on another organization's contribution, under which terms, or what must change when that contribution is withdrawn. We propose Kyoei, a federated layer on Git. Each organization's node signs its contributions and releases them to named recipients for stated purposes and a limited time. When an owner withdraws or corrects a contribution, each node blocks its own dependent work, and people approve the repairs. These checks cover declared dependencies and uses that pass through a cooperating node; they neither recall fetched copies nor establish correctness. We implement a research prototype and a Cloudflare port. In simulations that we designed, Git and a shared folder left 8 of 42 decisions to people. Kyoei's prototype and an equipped reference that we built each held all 66 of our fault trials. Adopting the prototype required no application code, against 2,470 lines for building the reference. However, a second collaboration added 363 lines to Kyoei itself against 360 new lines for the reference; we did not measure human effort or adoption. In one run in a single Cloudflare account, 16 agent runs completed, with six overlapping logged lifetimes. The integrator flagged two agents of one organization that declared the same file before either finished. One withdrawal blocked one line of work at each of three organizations. A script acting for their people selected released alternatives or replacements, and repair agents produced new versions. The ship gate checked signatures, release terms and merge bases; no build or test of the generated code ran. This run exercises coordination on Workers and Artifacts; it does not establish reliable recovery or an advantage over another platform.

KEY WORDS

Version control; multi-agent software engineering; provenance; usage control; withdrawal and correction.

1 Introduction

We describe the context and goal of collaboration across organizations. We then use a request-signing example to explain the requirements. Finally, we present Kyoei's approach, implementations and evaluation scope.

1.1 Context and goal

Organizations combine software and research from their own teams and external contributors. People and AI agents work in parallel, while teams remain responsible for the results they accept. They need the evidence and terms for each input, and a response when its owner corrects or withdraws it.

We want people and AI agents to build shared results while each organization retains authority over its contributions. Kyoei means mutual prosperity in Japanese and names this vision of human and AI symbiosis. We propose a federated collaboration layer for evidence-backed contributions. In this design, organizations control their own nodes, keys, and decisions while exchanging versioned results. Our technical goal is to help teams identify affected work, check permitted use, and approve repairs.

Git records versions and declared authorship; code hosts add review and merging. Registries and advisories identify withdrawn or vulnerable package versions. Teams must still connect those records to their declared dependencies, permitted uses and repair decisions across organizations. Kyoei makes those connections in a shared layer; a team can also build them itself, as we did for comparison (section 5.4). The design extends versioned collaboration from files to claims, supporting evidence, and declared dependencies. It retains Git for code and history. Federation describes the intended authority model: our Cloudflare experiment uses one account and one operator, so it does not establish independent administration or provider independence.

1.2 A request-signing example

Consider three organizations developing a client library for network requests. Birch (B) supplies a function that signs requests with a shared secret. Acme (A) builds a helper that retries failed requests and calls Birch's signing function. Cove (C) integrates their contributions with its own client code. Each organization's agents work in separate repositories.

A *claim* is a statement about a result that a recipient can assess and build on. Birch could claim that its function produces the specified signature for a given request and secret. A matching test output would supply evidence. The code version, producer, rationale and declared inputs describe the result's provenance. A *contribution* is the signed record of a claim and its provenance. The organization's signature binds that record to a key; it does not establish the claim's truth.

Birch then discovers a snippet whose licence it cannot establish and withdraws the signer. Under the collaboration's rules, Acme's retry helper and Cove's dependent client code must wait for repair before they can ship. Birch's documentation, which merely cites the signer, needs a notice. An unrelated README update at Cove remains eligible to ship.

Cove has also released an alternative signing function to Acme. Acme can adapt its helper to that alternative instead of waiting for Birch's replacement. The revised helper needs fresh review and approval. This example shows that a clean Git merge cannot decide whether the helper may still ship. That decision needs a record of the exact dependency and of its permitted use.

1.3 Requirements

The example shows five requirements for sharing and using a claim across organizations:

1. **Context.** Record the task, rationale and exact inputs of each change, so a recipient can inspect its basis.
2. **Terms.** State the recipients, purposes and expiry of a release, and check those terms when a recipient uses it.
3. **Concurrency.** Expose overlaps while agents work, then check both the merge base and the inputs before accepting a change.
4. **Change after use.** Follow declared dependencies after a withdrawal or correction, identify work that needs repair, and keep unrelated work usable.
5. **Authority.** Let each organization control its keys, records and decisions, with agents acting under delegation from its people.

1.4 Approach and scope

We specify a protocol in which each organization runs a node that records signed contributions and scoped releases. Nodes exchange change notices and compute their own affected work from declared dependencies. People choose and approve repairs. Section 5.8 reports the gaps that remain in the port, including exports for repositories with undisclosed history, personal signing keys and fault tests on a deployment.

We built two implementations. The TypeScript port runs one Cloudflare Worker, Durable Object and Artifacts namespace per organization, with a repository for each agent run. The earlier Python research prototype runs separate institution processes with their own stores and signing keys. It implements releases, purpose checks and repair, but lacks intents, git notes and the ship gate; several rules also differ from the protocol specified here.

We compare the research prototype with plain tools and an *equipped reference* that we built from the same libraries to provide the same controls. We also record deployed Cloudflare runs. The organizations and people are simulated, and we wrote the scenarios; these experiments do not establish savings for real teams.

Cloudflare asks how concurrent agents coordinate work, retain context and decide what ships.[1] Table 1 relates the port's mechanisms to that call.

Table 1: Cloudflare's questions and the mechanism in the Kyoei port that answers each.

Question or theme in the call[1]	Kyoei mechanism	Section
How do agents know what other agents are working on?	Declared intents show people when two running agents plan to change the same file; the agents are not warned.	4.2
What happens when they make conflicting changes?	The gate refuses stale bases and blocked inputs. A repair needs fresh approval.	4.5, 4.6
How do you review everything they produce?	Approval binds an exact version and its inputs. The gate displays each check.	4.5
How do you keep track of not just what changed, but why a change was made?	A git note carries the signed task, rationale, inputs, model, and prompt digest.	4.2
Compare several changes and decide which one should ship (a theme of the call)	The gate checks eligibility; it does not rank alternatives.	4.5

Kyoei controls *managed use*: operations that pass through a node's checks. A recipient that modifies its own node can bypass those checks, and fetched copies remain outside them. People must assess correctness, licence validity and ownership. Kyoei records their decisions; it cannot find a dependency that no contributor declared.

2 State of the art and requirements

In this section, we review version control and code hosting, package registries, and agent workspaces. We then cover open research platforms, provenance records, usage control, and the handling of withdrawn research. Finally, we compare what these tools provide with what Kyoei adds.

2.1 Version control and code hosting

Git stores content-addressed history. A commit identifies its exact tree and parents, and it can carry a signature.[2] Author and committer fields are declarations: a verified signature binds a commit to a key, and attributing that key to a person needs a separate trust decision. Git notes attach text to a commit under a separate reference without changing the commit.[3] Code hosts add pull requests, review, branch protection and merge queues. GitHub, for example, can give outside collaborators access to one repository[4] and builds a dependency graph from package manifests.[5]

These tools answer "what changed", "who changed it" and "who may write here". A GitHub personal access token can expire and carry a note about its purpose,[6] but GitHub does not check that purpose when someone uses the token. Access also does not follow a contribution once another organization fetches it. A dependency graph tracks declared packages, not which change relies on another organization's contribution. A team that needs that record, and a response when the contribution changes, must build an integration, as our equipped reference does.

2.2 Package registries and advisories

Package registries let an owner mark a published version so that new users avoid it, and the effect differs by registry. Go marks a version as retracted in the module's own file, so the go command stops upgrading to it but keeps existing builds working.[7] PyPI[8] and Cargo[9] let an owner yank a release; a pinned requirement or a lock file can still use it. npm prints a warning at each install of a deprecated version.[10] A published GitHub advisory can alert the repositories that depend on an affected version,[11] and Dependabot can open a pull request that updates the dependency.[12]

These mechanisms change version selection or warn users, and some keep pinned versions usable. Kyoei instead refuses new managed use once a node observes a withdrawal. It adds typed reliance records and repair decisions that each organization makes for its own work. Private registries can also restrict who downloads a version, so Kyoei differs from them in its response to a withdrawal, not in access control.

2.3 Agent workspaces and concurrency

Coding agents work in isolated workspaces: worktrees, sandboxes or forks. In a plain Git workflow, their conflicts appear at merge time. Research tools report them earlier. Palantír shows each developer which artifacts other workspaces are changing,[13] and Crystal merges parallel changes in the background to report conflicts before anyone merges.[14] Kyoei's declared intents follow Palantír's approach across organizations: the integrator's node compares the files that running agents plan to change (section 4.2). Kyoei does not merge changes in the background as Crystal does. Cloudflare Artifacts lets a Worker create and fork Git repositories, mint repository-scoped tokens and subscribe to repository events.[15] Multi-agent frameworks such as AutoGen structure conversations between agents, tools and people;[16] a deployment must still define release authority and the response to changes across organizations.

We distinguish three kinds of conflict. A *text conflict* occurs when two changes edit the same lines; Git detects it at merge. A *file overlap* occurs when two changes edit different lines of the same file: Git merges them, although neither change was reviewed with the other. A *dependency conflict* occurs when a change merges cleanly but relies on an input that is no longer valid; a text merge cannot detect it. Kyoei and the equipped reference both check declared dependencies alongside Git.

2.4 Open research platforms

Open research platforms let anyone contribute to a shared problem under an automatic check. Yukon, for example, runs public challenges in which a verifier checks each submission and a harness scores it; a candidate that clears an improvement threshold is promoted.[17] The platform owns the official record of a challenge, its scoring and its corrections.

We replayed the public records of two such challenges.[18] The lineage of a promoted result spans a median of 12 declared models in one challenge and 21 in the other. One model field per submission therefore does not describe what a result builds on. When a challenge's checker changed, the earlier outputs that it had scored needed review. Over three checker changes, Git-only methods missed 1 of 406 such outputs, so inside one public repository, dependency tracking added little over Git. Kyoei's research prototype imports such records as labelled inputs and never changes them; the port has no

imports. Kyoei targets the work that continues after a result, such as joint write-ups, reproductions and follow-up projects across institutions.

2.5 Provenance, licence and supply-chain records

W3C PROV describes entities, activities and agents, and how one entity was derived from another.[19] The Citation Typing Ontology types citations, for example "uses method in" or "disputes".[20] SPDX identifies licences.[21] The supply-chain frameworks SLSA[22] and in-toto[23] record how an artifact was built and by whom. These records describe origin. None of them carries a grant for a stated purpose that a recipient checks at each use, and none starts a repair in another organization when a source changes. One store of Kyoei's research prototype maps its relations to PROV and the Citation Typing Ontology; the port has no such export.

2.6 Usage control and authorization

Usage control models allow continuous authorization: a decision can change while a use is ongoing, so a revoked right stops a use that has already started.[24] Zanzibar applies changes to permissions and content in order, so a user removed from a folder does not see content added after the removal.[25] Leases bound how long a client may use cached data without asking the server again.[26] ODRL expresses permissions, prohibitions and duties, and a rule can name the purpose of a use.[27] Macaroons attach caveats such as an expiry to a bearer token; any holder can add a caveat but cannot remove one.[28]

Kyoei uses four of these ideas to check releases between organizations. A release names its recipients and purposes, as an ODRL rule can, and an expiry that bounds their use, as a lease does. An onward release cannot exceed the grants of its inputs, just as a macaroon holder can add caveats but cannot remove them (section 4.3). As in Zanzibar, order matters: a node keeps the newest owner state, so a delayed release cannot clear a withdrawal that the node has already observed (section 4.4). Unlike a usage control model, a node checks a release before each managed use but does not stop one in progress.

2.7 Withdrawal and correction in research

Retraction notices, retraction databases and notification services alert readers after the fact. The RetractoBot trial sent 246,749 one-time emails to authors who had cited retracted articles. Of 15,667 survey responses from notified authors, 80.6% said that they had not known about the retractions. Over the following year, the emailed group's retracted articles were not cited significantly less often than those in the control group.[29] Usman and Balke separate dependent citations, which build on the retracted work, from non-dependent ones, which discuss, criticize or negate it.[30] In a case study by Bar-Ilan and Halevi, 83% of 238 citations made after a retraction cited the retracted work as legitimate prior work.[31] Kyoei therefore checks at the point of use rather than after the fact. Like Usman and Balke, it separates dependent from non-dependent links: a withdrawal blocks work that builds on the withdrawn result, and work that only cites it receives a notice.

2.8 Requirements and the Kyoei contribution

No single Kyoei mechanism is new. We combine four of them at the boundary between organizations, on top of Git: signed contribution records with typed relations; releases that the recipient's node checks at each managed use; affected work that each node computes from its own declared relations; and repair under human approval. Our equipped reference shows that a team can build the same combination itself. We contribute it as a shared layer, so that each team need not build it:

- a protocol specification for signed contributions, scoped releases, change events and repairs between organizations (section 4);

- a Python research prototype and a TypeScript port on Cloudflare Workers, Durable Objects and Artifacts (sections 5.1 and 5.8);
- an evaluation against plain tools and the equipped reference in three simulated collaborations, and two deployed runs on Cloudflare (section 5).

We did not measure whether the shared layer saves real teams any effort.

Table 2: Five requirements, existing tools, and the Kyoei design. The design column states goals, not complete port guarantees; section 5.8 lists the port's gaps.

Requirement	Existing tools provide	Kyoei design	Section
1 Context	Commit history, signatures, notes, pull request text	A signed contribution record with task, rationale, typed inputs, model and prompt digest, in the owner's ledger; the port also stores it as a git note	3.2, 4.2
2 Terms	Repository access per role; tokens with an expiry; licence identifiers	Releases for named organizations, stated purposes and an expiry, checked at each managed use	3.3, 4.3, 4.5
3 Concurrency	Isolated workspaces; text conflicts at merge	In the port: declared intents and early overlap signals; file overlaps and dependency conflicts at the ship gate	4.2, 4.5
4 Change after use	Reverts, issues, messages; yanked or retracted package versions; advisories that alert dependents; notices after the fact	Signed withdrawal and correction events; each node computes its own affected work; repair with human approval	4.6, 4.7
5 Authority	Accounts and roles on one host	One node, key and namespace per organization; only people release, approve and withdraw	3.1, 4.1, 4.3

3 Model and definitions

In this section, we define the participants and the collaboration cycle. We then define contributions and their relations, and releases and the managed use they authorize. Next, we define the changes that start a repair, the work they affect and the notices they open. We close with refusals, repairs and the scope of a collaboration.

3.1 Participants and the collaboration cycle

An *organization* controls its people, agents, keys, repositories and decisions. In the request-signing example, Birch and Acme contribute and Cove integrates; any organization can take both roles. Each organization runs one Kyoei *node*, which keeps the organization's ledger and signing key, pins the public keys of its peers, and sends its outgoing events. An *actor* is a person or an agent of one organization. Agents produce work under a *delegation* from a person, which names the allowed tasks, files, purposes

and expiry. Only a person releases a contribution, approves or ships it, withdraws or challenges it, records dissent or chooses a repair.

Organizations join a collaboration by pinning each other's keys. Figure 1 summarizes the specified workflow cycle, not a conformance result. Its six stages follow one contribution, and section 4 specifies each stage.

In *work and context*, people or agents edit a separate repository, and the node signs a record of the version, its context and its declared inputs. In *release*, a person selects the recipients, purposes and expiry, and the node sends them the terms, the record and a read token. In *rely and build*, the recipient's node checks the released content, an agent there declares reliance on it, and the owner records which organizations rely on its work. At the *ship gate*, a person at the integrator approves the version, and the integrator's node merges it only if every check passes.

A released input can change at any stage. In *change and notices*, nodes send signed events, and each node finds its own affected work from the inputs declared in the records it holds. The node then blocks managed use of work that builds on a withdrawn or corrected input. It opens a notice for work that cites or challenges the input, and it keeps unrelated work usable under its own gate checks. In *repair*, a person chooses a new input, an agent or a person edits the work, and the node keeps the old record. The repair is a new version, so it passes the release and approval checks again; it does not restore the old version.

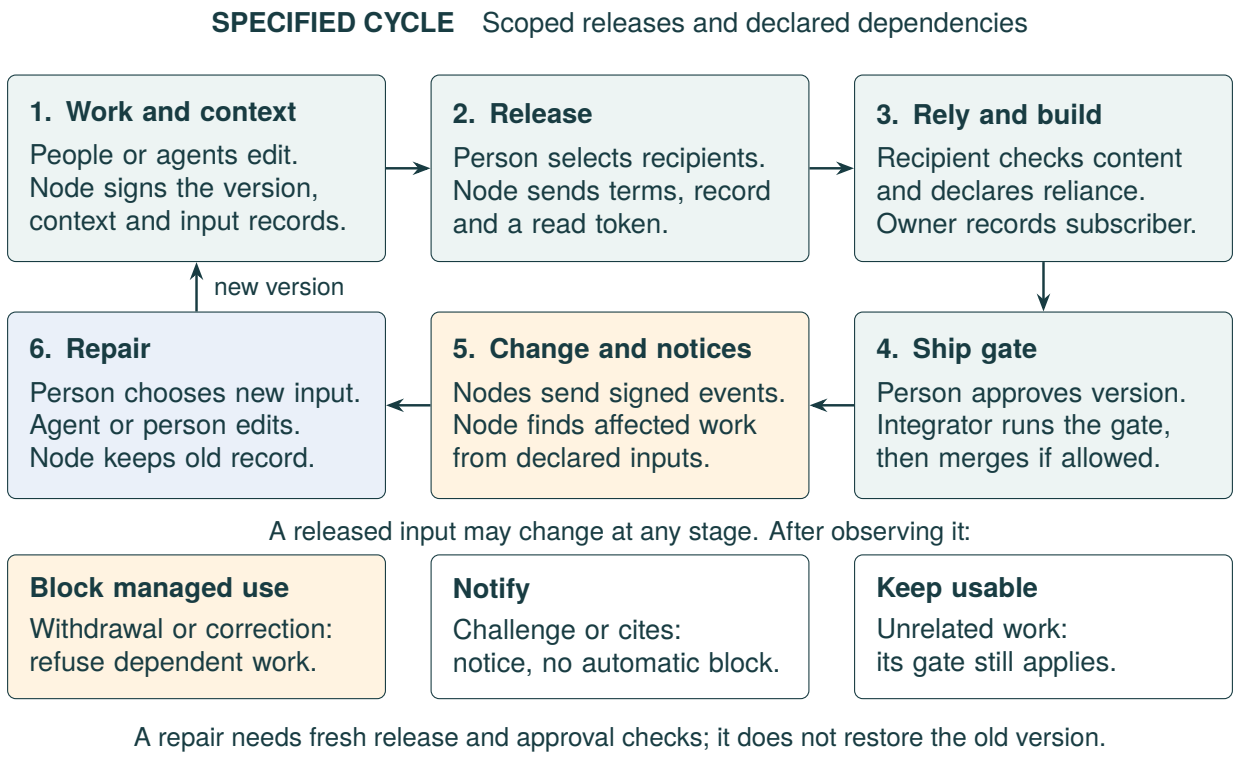


Figure 1: Workflow cycle under the specified protocol. A person chooses a replacement from the same owner or an authorized alternative from another organization. Checks apply to managed use; fetched copies remain outside them. Section 5.8 lists implementation gaps.

3.2 Contributions and relations

A *contribution* is the signed record of one claim. The record binds the result's commit and its *content root*, a hash over the committed files (section 4.3), and with them any evidence committed with the result. It names the organization, actor, task, rationale and declared inputs, and the interfaces that the result provides, such as `signer@1`; agent records also name the model and prompt digest. The record states no terms: releases carry them.

An *input* is a typed relation from the contribution to a versioned target. The type decides what a node does with the contribution when the target changes:

- *builds-on*: the contribution depends on the target. A withdrawal or correction of the target blocks the contribution until a person repairs it; a challenge to the target opens a notice. No other relation blocks.
- *cites*: the contribution mentions the target. A change to the target produces only a notice.
- *challenges*: the contribution disputes the target. The disagreement stays visible, and any change produces a notice.
- *reproduces*: the contribution repeats the target's result. A change to the target neither blocks the contribution nor opens a notice; the protocol does not yet define a response for this relation.
- *supersedes*: a contribution of the same owner replaces the target, as a correction does (section 3.4).

A *challenges* relation records disagreement inside one's own contribution; a *challenge* is a separate signed statement (section 3.4). The port registers each input except a supersedes input with its owner, so the owner's change events reach every organization that declared one of the other relations. The research prototype uses different blocking rules (section 5.1).

3.3 Releases and managed use

A *release* names one contribution by its record digest, the recipient organizations, the purposes and an expiry, and the owning organization signs it. *Managed use* is an operation that passes through the recipient's node, which checks that a release covers it. In the current port, the ship gate checks the releases of the candidate and of its declared builds-on inputs for the purpose `integrate`. These checks include expiry and completed reliance registration (section 5.8). The gate decides from the latest owner state that its node has authenticated; it does not query each owner at each use. A withdrawal that has not yet arrived therefore does not stop the use (section 4.6). In the port, a release also carries a read token for one repository, which Artifacts checks for scope and expiry on a Git fetch. That bearer token neither enforces the recipient's stated purpose nor prevents onward copying, and a raw fetch runs none of the node's checks, even though Kyoei issued the token. Repository access is therefore distinct from approved reuse. In the research prototype, managed use is a retrieval, use, publication or release through the node. In both, a copy already fetched is outside managed use.

3.4 Changes, affected work and notices

A *withdrawal* ends the owner's releases and endorsement of a contribution; it does not rescind licences already granted. A *correction* is a new contribution that supersedes an old one. A *challenge* is a signed statement by a person of any member organization that a contribution is wrong; it does not block by itself. *Dissent* is a recorded disagreement with a notice or a challenge; it does not clear a withdrawal, grant permission or restore eligibility to ship.

Let O_N be the contributions that node N owns, and let E_N be the relations declared by the records that N holds: its own and those released to it. For a contribution c that changed, was blocked or was challenged, the *affected set* of N is

$$A_N(c) = \{x \in O_N : x \xrightarrow{\text{builds-on}^+} c \text{ in } E_N\},$$

the contributions of N that reach c through one or more builds-on relations in E_N. A node that is not a recipient of c learns of a change through `blocked` events and of a challenge through `affected` events (sections 4.6 and 4.7). The state of c itself is checked separately; it needs no relation to itself to be refused. The *notice-only set* $M_N(c)$ holds the contributions of N that cite or challenge c and are not in $A_N(c)$. Each node computes both sets itself.

A *notice* is a node's record that one of its contributions is in $A_N(c)$ or $M_N(c)$. It has six fields: the affected item; a reference to its version, which the port gives as the commit, record digest and content root; the source event; who can act; any dissent; and the allowed repairs.

3.5 Refusals, repairs and collaboration scope

In the port, a refused managed use carries one of seven codes: `withdrawn`, `replacement-required`, `new-grant-required`, `lease-expired` for an expired release or envelope, `human-required`, `stale-base` and `not-authorized`. A malformed request returns `invalid-request`, and an event identifier reused with another payload returns `conflict`. The node's HTTP interface also returns `not-found` for an unknown object and `unavailable` for any other error. A *repair* creates a new version with revised inputs or a current base, and it ships only after fresh approval and all gate checks pass. The owner of blocked work can instead withdraw it, which ends its use without making it shippable.

A *collaboration* is the scope in which releases, relations and notices apply. A node refuses a relation into another collaboration, except to a labelled import, and it does not disclose the state of another collaboration's work. An import needs an explicit grant for its new scope; the label alone neither authorizes access nor exposes the source collaboration's private lineage. Only the research prototype implements imports; the port runs one collaboration per node. A signed record names its deployment, collaboration and owner under a versioned signing tag, and a receiver checks them against the envelope and its membership records. A valid signature under an organization key therefore does not make a record usable in another collaboration that trusts the same key; moving the record there needs an authorized import with a new record identity.

4 The Kyoei protocol

In this section, we specify the protocol; section 5.8 reports which parts the Cloudflare port implements and tests. We describe setup, signed envelopes and their delivery. We then follow one change through work, release, reliance and the ship gate. We describe withdrawal, correction and repair, then challenges and dissent. Finally, we give the threat model and the evidence for each property.

4.1 Setup, envelopes and delivery

Each organization creates a signing key. The research prototype and the deployed port sign with ML-DSA-65, a post-quantum signature standard.[32] The port's local tests and offline demonstration sign with Ed25519. Organizations exchange key fingerprints out of band and pin them. On Cloudflare, each organization deploys the same Worker code with its own Durable Objects,[33] Artifacts namespace[15] and secrets, so one organization's bindings cannot reach another organization's storage or repositories.

Every message between nodes is a signed envelope. An envelope names its event identifier, issuer, audience, kind, collaboration, payload, issue time, expiry and nonce. The signature covers a canonical encoding of all fields, including a protocol version and a deployment identifier. In the port, an envelope lives 10 minutes. A receiver refuses an unknown issuer, an invalid signature, an expired envelope, and a wrong audience, deployment or collaboration. It also checks authority over the named object: a pinned key alone cannot withdraw another owner's contribution, and only the owner of the old record may issue `correct`. The event kinds are `release`, `rely`, `intent`, `overlap`, `withdraw`, `correct`, `blocked`, `affected`, `challenge`, `dissent` and `signoff`.

Delivery needs more than a valid signature. Each event keeps a stable identifier across retries, and a durable inbox applies it once: a retry returns the existing receipt, and reusing the identifier with another payload is refused. An event retried after its envelope expired is re-sealed with the same identifier and payload. A receipt means that the recipient committed the effect, its inbox record and any events that the effect caused. The sender keeps each outbox entry until that receipt arrives, and after a crash it reclaims and resends the entries that it had claimed. A permanent refusal stays in the sender's outbox

with its error and appears in its log. These rules assume that storage is not restored to an older snapshot and that node clocks differ by less than the 120 seconds of skew that a receiver allows. Signatures do not encrypt, so envelopes travel over HTTPS to each peer's recorded address; the nodes pin each other's signing keys, not TLS certificates. Bearer tokens never appear in git notes, logs or error messages.

4.2 Work and context

Before an agent run starts, its node creates a repository for the run from the head of the integration repository and mints a write token scoped to that repository. The run declares an *intent*: its task and the files it may change. Each node sends the intents of its runs to the integrator as signed `intent` events and closes each one when the run's contribution is recorded. The integrator's node compares the open intents of every organization's agents by exact file path and sends an `overlap` event to each organization involved. People see the overlap in the terminal view while the work runs, and agents are not told; an overlap warns and does not block. The model returns its edit as JSON, and the node accepts only files inside the run's file list and under a size limit. The node, not the model, holds the write token. It commits the edit and writes the contribution record as a git note under `refs/notes/kyoei`: the task; the rationale; the inputs and their relations; and the model and prompt digest. The node signs the record and stores it in its ledger. A client that fetches `refs/notes/kyoei` can read the note with `git log --notes=kyoei`; a plain clone does not fetch notes.

Releases, approvals, and relations identify a record by the digest of its fields except the signature. That digest covers the commit identifier and every other field, so two records about one commit can differ, and a changed record needs a fresh signature and approval. Notes themselves give no authority because they can change without changing the commit.[3]

4.3 Release

A person releases a contribution, and the organization's own node refuses a release by an agent with `human-required`. Peers see the organization's signature, not the person. The node mints a read token for the contribution's repository and sends a signed `release` envelope to each recipient. The envelope names the contribution, the purposes and the expiry, and it carries the token. The token expires with the release, unless the minimum lifetime of an Artifacts token is longer (section 5.8).

Git object names depend on the repository's hash format, SHA-1 by default. The record therefore also binds the released files with a hash that does not depend on that format. Its content field carries a BLAKE3 root computed over the canonical path, the Git mode and the raw bytes of each committed file. A recipient checks the record's signature and expected scope, then recomputes the root from the exact commit that it fetched. The root authenticates the committed files under the hash and signature assumptions. It does not authenticate parent commits, tags or notes, and it proves neither correctness, permission nor secrecy. Symlink targets are hashed as bytes without being followed. Submodules and paths with an empty, `.`, `..` or `.git` segment are refused, but paths that differ only in letter case are accepted. On a file system that ignores letter case, a checkout of two such paths keeps only one file. A reviewer can then see different files from those that the root covers.

A release carries the full signed record, and its read token opens the whole repository and its history, so no release selects which evidence or inputs a recipient sees. If the files or history hold unreleased material, as a fork's history can, the owner must first publish a separate export repository. That export helps only if no other authorized repository exposes the same version. Revoking a token removes only that credential. If the release of another contribution grants a token for the same repository, the recipient can still fetch the withdrawn version with that token. The Artifacts documentation also does not say whether a revocation stops a clone already in progress.

Ownership of a derived contribution grants no permission over its inputs. Consider a signer that Birch releases only to Acme. Acme then releases to Cove a helper that builds on that signer. Acme's signature alone cannot authorize Cove to build on Birch's signer through the helper. Acme's node therefore refuses the release unless Birch's grant to Acme allows release to Cove for every proposed purpose,

and it ends the release no later than Birch's grant. The node checks only builds-on inputs, infers no permission from a builds-on link and does not interpret licences.

4.4 Building on released work

An agent in a recipient organization declares a relation to a released contribution. The recipient's node checks that a release for building on the contribution is in force, with the purpose `contribute` in the port, and records the relation. It then sends a signed `rely` request to the owner and waits for the reply. The owner records a *reliance*: which organization declared a relation to which of its contributions, whatever the relation type. If the owner released the contribution to that organization, it also registers the organization as a *subscriber* of the contribution, which then receives its change events.

A new reliance can arrive after a withdrawal. The owner therefore reads its current state and registers the subscriber in one serialized step, and returns that state in its reply; otherwise a withdrawal between the two steps could reach neither the reply nor the subscriber. The reply binds the request, the record, the scope and the owner's state revision. The receiver keeps the newest revision, and a delayed release never clears an observed withdrawal. Unknown state leaves the reliance pending, and the gate refuses dependent work with `not-authorized` and the reason "owner state unknown". A node shows missing state, failed delivery and unresolved inputs as unresolved; it never shows such a contribution as clear to use. The serialized step closes the race between registration and withdrawal at the owner, under the delivery rules of section 4.1; it does not make the reply current at every node.

4.5 The ship gate

Before the integrator merges a contribution, its gate checks six conditions:

1. the record's signature, which the node verified when the record arrived;
2. the content root;
3. a release to the integrator that covers the purpose `integrate`, with a completed reliance registration when another organization owns the contribution;
4. the builds-on inputs: none may have an unresolved withdrawal or correction or a pending registration, and their releases must still cover the use;
5. a person's approval;
6. a base that is current for the candidate's files.

The base is current when it is on the main line and no later merge changed a file that the candidate also changes. A change on a stale base therefore returns `stale-base` even when its text does not conflict. The gate shows each check and its reason.

The gate does not build or test the candidate. Checks of behaviour remain with the team's own build and tests; the port neither runs nor records them.

Approval binds the candidate commit, the contribution record, its content root, its inputs and the base that the record declares, so a changed diff or input needs fresh approval. When a person approves a contribution that another organization owns, the approving node sends a `signoff` event to the owner. The owner stores it; no check depends on it, and it names neither the commit nor the content root.

The gate's decision and the Artifacts ref update are separate effects, and no transaction spans them. The node therefore conditions the update on the reviewed base, keeps an unknown Git outcome unresolved rather than failed, and reconciles it against the actual ref and its note. The gate authorizes a merge at its decision. A withdrawal observed after that decision does not undo the merge; it blocks further managed use of the merged item and starts a repair.

4.6 Withdrawal, correction and repair

A person at the owning organization withdraws a contribution. The owner's node revokes every read token that it issued for that contribution and sends a signed `withdraw` event to every recipient and subscriber. Each receiving node computes $A_N(c)$ and $M_N(c)$ over the records it holds. It blocks managed use of $A_N(c)$ with `withdrawn`, opens a notice for each item in both sets, and leaves every other item unchanged. If a withdrawal arrives before its release, the node keeps a record of the withdrawal, so the late release arrives already withdrawn.

Blocked work can have dependents in a third organization. When a node blocks one of its own contributions, it sends a signed `blocked` event to each organization that declared reliance on that contribution. Each receiving node checks that the sender owns the named input version, then computes its own affected work. Status thus follows declared relations, and no node sees more of the graph than the records it holds. Until the `withdraw` or `blocked` event arrives, a recipient can still allow managed use of a released item, but no longer than its release. Every release expires, and no onward release outlasts the grants of its builds-on inputs (section 4.3).

A correction creates a new contribution that declares `supersedes c`. The owner sends `correct` to existing recipients and subscribers, even if the new version is not released to them. A recipient marks the old version as `replacement-required` and propagates the effect to its dependents. It gains no access to the replacement, and an earlier withdrawal stays in force. A recipient can also select an alternative: an unblocked contribution that its node may build on and that provides an interface of the blocked input, such as `signer@1`. The node lists the alternatives and the owner's replacement, if one exists, and never applies either silently. An alternative from another organization does not supersede the original owner's record.

A repair keeps the old blocked version for traceability. If two inputs are withdrawn, a repair that replaces only one stays blocked, and the port refuses to plan it. A matching interface name does not establish a valid replacement, and a clean rebase establishes neither equivalent behaviour nor correctness. The reviewer must therefore assess the new input and its evidence for the dependent's task.

4.7 Challenges and dissent

A person at any member organization can challenge a contribution. The challenge opens notices for the work that builds on or cites the challenged contribution, and each notice names the challenge as its source. The owner can withdraw or correct the contribution, or record dissent; the integrator can also decide not to ship. Each response stays in the record.

A notice must also reach organizations that rely on an affected result. Suppose Acme's helper builds on Birch's signer, and Cove's integration builds on Acme's helper. A challenge to the signer does not block the helper, so forwarding only `blocked` events would leave Cove unaware. Acme's node therefore sends an `affected` event for its helper, bound to the helper's version and a stable cause identifier. Cove checks Acme's authority over that helper and computes notices from its own declared relations. The `affected` event neither lets Acme withdraw Birch's signer nor lets Cove treat the challenge as a proven defect. A citation-only notice does not propagate, and nodes deduplicate each cause and version. An `affected` event carries only the challenge evidence that its recipient is allowed to see, and it marks withheld evidence as withheld. Cove can then verify Acme's statement about the helper but cannot assess a hidden challenge.

Dissent goes to the organization that issued the event behind the notice, such as the challenger, not to every organization that received the notice. An organization that only forwarded the notice does not receive the dissent.

4.8 Threat model and properties

Each organization's node enforces the protocol for that organization's own people and agents. We therefore assume that each organization runs an unmodified node. Under that assumption, we consider five threats: a party that forges or replays events; a person or agent that tries to use work beyond its terms; an agent that exceeds its delegation; material from another organization that contains instructions for an agent; and a node that leaks the state of another collaboration. The protocol targets seven properties. They are requirements, not theorems, and Table 3 gives the evidence for each in each implementation.

- **P1 Local authority.** No node writes another node's records. Each node computes its own affected set.
- **P2 Human decisions.** An agent cannot release, approve, ship, withdraw or challenge a contribution, record dissent or choose a repair.
- **P3 Dependency scope.** Once an input is blocked, downstream blocking follows only builds-on relations. Its owner's withdrawal also blocks the input itself.
- **P4 Reach.** Every recipient of a changed contribution, and every organization that declared reliance on a blocked or affected one, receives an event.
- **P5 Integrity.** Forged, expired and misaddressed events are refused, and a replayed event has no second effect.
- **P6 Isolation.** A node refuses relations into another collaboration and hides the state of another collaboration's work, except for explicitly authorized imports and their permitted disclosures.
- **P7 Terms.** A node refuses managed use that no release in force covers by recipient, purpose and expiry, and an onward release beyond the grants of its inputs.

P3 and P4 apply only to *declared* relations: no node can find a dependency that nobody declared. P4 also assumes that each node on the path cooperates, keeps its outbox and retries each event until it is acknowledged; no finite delivery deadline follows. P2 rests on actor roles checked by the local node. In the prototype runs, one parent process generated every key. In the port, one administrator token per node authorizes every person action. The request names the acting person, so the token's holder can act as any of that node's people. Neither implementation establishes independent key custody or independent human approval.

Table 3: Who enforces each property, and the evidence for it in each implementation. Port test names refer to the port's acceptance tests (section 5.8); "weak" marks a test that asserts less than its name states.

Property	Enforced by	Research prototype	Cloudflare port
P1 Local authority	Each node, for its own records	Withdrawal run: every request signed by the acting institution; no process held another institution's store	Separate Worker deployments with no shared binding
P2 Human decisions	The acting organization's node	Fault class "delegation escalation": an agent's release refused with `human-required`	R01 (weak) and R13; no deployed run had an agent attempt a person-only action
P3 Dependency scope	Each node, over the records it holds	Diagnostic notices match declared reliance; refusal at every new use after a block is not established	R06 (weak), R08 and R11; the deployed run blocked only dependent lines
P4 Reach	The owner's node, then each node that forwards `blocked`	One loopback run delivered the expected notices; no partition, restart or expiry test	R14 (weak), R18, R22, R23 and local crash-recovery tests R26, R27, R37 and R39 (section 5.8)

Property	Enforced by	Research prototype	Cloudflare port
P5 Integrity	The receiving node	Fault classes for forgery, expiry, replay, reordered status and key rotation, under the pass rule of section 5.2	R02 refuses nine kinds of bad envelope; R23
P6 Isolation	Each node	Third-collaboration scans and base-store tests; no information-flow or metadata-privacy proof	R03 for record scope; one collaboration per node, with no imports
P7 Terms	The recipient's node; the releasing node for onward releases	Fault classes for a new purpose without a grant, prohibited disclosure and an unreleased draft input, under the pass rule of section 5.2	R05, R19 (weak), R24 and R25; the current gate checks input grants and pending candidate reliance (section 5.8)

Tests cover parts of these properties; none is proved end to end, no part of the protocol is mechanically verified, and no external security review has taken place. Several threats remain open. Whoever controls a node's cloud account can read its signing key and ledger. An owner can send different events to different recipients, and nodes do not compare what they received. A transparency log, as in Certificate Transparency,[34] could support detection only with inclusion checks, monitoring and comparison of accepted histories. A stolen node key lets its holder sign as that organization until its peers pin a new key. Material from another organization can still steer an agent's output inside the task's files; schemas, file lists, size limits and human approval limit the damage but do not prevent it. A participant can flood a collaboration with challenges, and a withdrawal stalls every dependent until someone repairs it.

Private contents can stay local while metadata crosses the boundary. A reliance event tells the owner which organization uses its contribution. An intent tells the integrator a run's task and files before any release, and a release discloses the full record with its inputs. Notices disclose identifiers, timing and some provenance. Digests of guessable material can also reveal information. No result here proves confidentiality of the collaboration graph.

5 Implementation and evaluation

In this section, we describe the research prototype and state the evaluation method. We report the plain-tools run, the equipped reference in a first collaboration, and reuse in later collaborations. We then report dependency proposals by agents and withdrawal across separate nodes. We close with the Cloudflare port, its tests and its known gaps, and with two deployed runs and their architecture.

5.1 The research prototype

The prototype is written in Python with the standard library and three pinned cryptography packages. In the collaboration runs and the withdrawal run, each institution runs as a separate process with its own SQLite store; the fault trials run all institutions in one process. The prototype calls organizations *institutions*, because it started with research collaborations. Table 4 lists how it implements each mechanism. Several mechanisms differ from section 4, and the results in sections 5.3 to 5.7 come from these mechanisms.

Table 4: How the research prototype implements each mechanism. "Collaborations" are the three collaboration runs; the "withdrawal run" is that of section 5.7.

Mechanism	Research prototype
Signatures	ML-DSA-65 from dilithium-py 1.4.0, on each client request with the actor's key and on each relay event with the institution's key
Transport	Unix-domain sockets in the collaborations; plain HTTP on the loopback interface in the withdrawal run; no TLS
Freshness and replay	A request expires at most 300 seconds after issue, and a replayed write returns its first receipt. A relay event expires after 60 seconds and has no audience or nonce; a duplicate has no effect
Release	A record at the owner. The recipient's person fetches a signed view, which withholds the work's own inputs
How a withdrawal reaches a recipient	At the recipient's next status query to the owner (collaborations); as a pushed event that is not resent after a refusal (withdrawal run)
Delivery recovery after a crash	None
Use while the owner is unreachable	Allowed until an owner-signed lease ends (collaborations); state reported, not checked at use (withdrawal run)
Relations that block	Every captured input, whatever its type, while a relation declared afterwards never blocks (collaborations); builds-on, reproduces and supersedes (withdrawal run)
Reach beyond direct recipients	A status query asks each input's owner in turn (collaborations); `affected` and `blocked` events forwarded along declared reliance (withdrawal run)
Correction	Dependents become correction-pending, then replacement-required once acknowledged; a withdrawn input stays withdrawn
Who may record a withdrawal	The owner, for its own work; any trusted institution, for an outside source (withdrawal run)
Intents, overlap signals, git notes, ship gate, content root	Not implemented
Refusal codes	Many specific codes, for example `correction-blocked`, `release-required` and `onward-release-required`

5.2 Method

The evaluation follows five rules.

1. **Freeze first.** The scenario, the expected outcomes, the counting rules and the predictions are frozen with a SHA-256 digest before any measured run. Each prediction has a falsifier that names the result refuting it. A local digest detects later changes; it does not independently timestamp the rules.
2. **Reference first.** An equipped reference, built from the same cryptographic libraries and in part from Kyoei's own modules, is qualified on the same fault schedule and measured before Kyoei.
3. **Plain tools.** The same scenario runs against Git and a shared folder, with the tools' rules simulated.
4. **Score outside the arm.** Scorers read stored traces, and drivers do not read the expected answers. This prevents direct scoring from an oracle; it does not make the fixture or its authors independent.
5. **Check cited measurements.** The paper's `verify.py` pairs each listed quantitative statement, and its restatements in the abstract and conclusion, with saved evidence or source text. A release also requires every cited evidence file to be committed. A passing check detects drift; it does not validate a metric or rule out bias. Results that failed review were withdrawn and are not used.

Six departures from these rules affect results in this paper. The rule that excludes throughput cells measured under outside load was frozen while the reference grid was at cell 23 of 80. It excluded

3 reference cells, and the same cells in every arm. The rules that map the frozen events of the third collaboration onto each system's operations were written after its scenario was frozen, before its measured runs. In the second collaboration, we revised how Kyoei's setup and repair steps are counted after seeing its first counts; the evidence keeps both counts. In the withdrawal run, we aligned the notice format with the expected answers after reading them. Only the withdrawal run follows rule 4: the drivers of the three collaborations compare each event with its expected outcome during the run. The Cloudflare runs follow none of rules 1 to 3: they have no frozen predictions, no reference arm and no plain-tools arm.

The measures count lines of code, configuration decisions, commands, steps, refusals and time. They do not measure human effort, usability or adoption.

The first collaboration (C1) has 42 events for three simulated institutions. It covers enrollment, contribution, release, delegation to agents with limits, a shared draft with a concurrent edit, correction, withdrawal, reuse for a new purpose, export at exit and public release. One event needs a selective disclosure proof, which no arm implements. The second collaboration (C2) has 50 events for the same institutions under new terms, and the third (C3) has 14 events under a new purpose.

The fault schedule has 22 classes: concurrent correction and use; delegation escalation; dependency corruption; duplicate message; expired message; forged actor; forged message; change of Git history; interrupted write; issuer outage; key rotation; licence conflict; lost push; malicious source; new purpose without a grant; use after an observed withdrawal; prohibited disclosure; reordered status; restart; stale draft base; unreleased draft input; and wrong owner. Each class runs three times per arm. A trial *holds* when the attack ends in a refusal or an error, the valid control succeeds, the refusal text holds no private marker and nothing crashes. The trials do not compare the refusal code with an expected code.

Five classes test the harness more than a node's own check. The harness itself determines the duplicate-message control and Kyoei's restart result. The Git-history class runs the same Git check in every arm. The malicious-source class checks for a file that no arm could create, because no arm runs an agent. In the issuer-outage class, the harness raises the connection error itself, so the attack never reaches a Kyoei node. Kyoei runs in two modes, with document sources and with Git sources; both run the same trials, so the Git mode adds no fault coverage. One accepted case is recorded: a use that starts before a correction is allowed, and only the next use is refused.

5.3 Plain tools

Table 5: The 42-event scenario with Git and a shared folder only (simulated tool rules).

Outcome	Git and a shared folder
Decisions that match the scenario	34 of 42
Withdrawn or outdated work accepted for use	5
Prohibited exchanges or public releases	2
Reuse for a new purpose without a grant	1
Withdrawn items left in the exit export	5
Lost updates in the concurrent draft edit	1 with a shared file; 0 with Git
Affected outputs flagged, dependency links, delegation limits	None stored; the scenario has 19 affected outputs, 32 links and 3 limits

Git refused the stale push, so it lost no update. The eight mismatched decisions are uses and exchanges that the scenario forbids; with these tools, a person must notice each one. We wrote the scenario to exercise these controls, so the count shows which decisions the tools leave to people, not how often real teams get them wrong. The last row is structural: as configured, the tools store no such record, although Git notes could hold one, as they do in the Cloudflare port.

5.4 An equipped reference in a first collaboration

Table 6: First collaboration (C1), equipped reference against Kyoei. The two Kyoei modes gave the same values except where noted; configuration and setup come from the document mode only. In the split throughput group, the two modes fell on opposite sides of the reference.

Measure	Reference	Kyoei	Note
Fault trials held	66 of 66	66 of 66 in each mode	22 classes, 3 trials each
Application code written by the adopting team	2,470 lines	none	The reference also reuses 2,091 lines of Kyoei's modules. Kyoei's own code is 3,684 lines, maintained by its supplier.
Dependency links recorded, of 32 expected	30, of which 20 automatic, plus 10 extra	32, of which 30 automatic, none extra	The reference's extra links tie joint outputs to their draft's inputs
Configuration decisions per institution	16	17	Keys read from code; their classification is the authors'
Setup commands	13	20	Unequal boundaries; see text
Requests slower than 5 s under load, 80 cells per arm	546	0	Includes 44 reference `concurrent-change` refusals
Refusals in original responses	953 `stale-base`; 44 `concurrent-change`	988 `stale-base` with documents; 986 with Git	Before latency classification
Throughput groups in which the arm was faster, of 16	5	10	One group was split

The fault schedule does not separate the two systems: both held every trial. The code count compares adopting a layer with building one. The reference's 2,470 lines are the amount we wrote under this reuse boundary, not a lower bound on an independent team's implementation. The reference's records missed 2 expected links and added 10 that tie joint outputs to their draft's inputs. Its correction and withdrawal sets still matched.

The setup totals use unequal boundaries. Both arms count three node starts, six trust commands and four enrollments. Kyoei also counts seven key initializations, one per actor. The reference generated the same seven keys inside its test harness, where they were not counted, so the whole setup difference is key provisioning. We keep the raw totals and do not turn the unmeasured provisioning into a corrected cost estimate.

All 546 slow reference requests occurred in mixed workloads with 27 actors. The harness counts every response slower than five seconds as a timeout, whatever its outcome: 44 of these requests ended in `concurrent-change`; the other 502 were allowed. The reported throughput counts allowed responses that meet the five-second threshold per second of measured wall time. It is not the completion rate of all requests. The slow-request totals include all 80 cells, and the faster-group comparison excludes the three contaminated cell keys from each arm. We ran the arms one after another on one shared machine, with no counterbalanced order and no replication.

We did not profile the slow requests. The reference stores each node's state in one JSON file that every write rewrites, a design that we chose, while Kyoei uses SQLite. The gap may reflect that choice rather than the protocol. Kyoei was faster in all 8 mixed-workload groups and in both read-only groups with 27 actors; the reference was faster in 5 read-only groups with 1 to 9 actors. The decision rule frozen

before the runs records a split across measures and states that these results neither validate nor disprove adoption.

5.5 Reuse in later collaborations

The same institutions then ran a second and a third collaboration on the same nodes. In the second collaboration (C2), the reference needed 360 new lines, and Kyoei's maintainers added 363 lines to Kyoei itself. The reference also needed fewer configuration decisions (21 against 22), setup commands (28 against 30) and repair steps (32 against 34). Before C2, we froze a test of the claim that reuse makes Kyoei cheaper. The claim fails if the reference needs no more new code, configuration decisions or setup commands than Kyoei, and matches it on repair and failures. Counting the 363 lines added to Kyoei as new code, the reference meets every condition, so the claim fails; counting only the adopter's code, none against 360, it survives. Kyoei's count passed 360 only after an 8-line isolation fix made after the first runs; before that fix, it was 355 lines. On every measure except adopter code, the second collaboration was not cheaper with Kyoei.

In the third collaboration (C3), under a new purpose, the reference needed a new 45-line adapter, because we had bound its stores to the earlier purpose. Kyoei's driver records no new code, but it does not check the supplier files added in the second collaboration. The reference can reuse that adapter for a later purpose, so the difference need not recur. Both arms allowed the same 20 scenario decisions and held both injected faults, three times each. The reference needed 20 client setup commands against 24 for Kyoei, plus three node restarts each; Kyoei needed 22 repair steps against 23 for the reference. A scan found no allowed request labelled with a second-collaboration item or B's private table, and no file at A or C with the first collaboration's private tables; it did not inspect the packages.

Across the three collaborations, these results do not establish lower total cost. Both arms could reuse prior code, Kyoei moved implementation work from the adopter to its supplier, and the second collaboration still required supplier changes.

5.6 Proposing dependencies with agents

Kyoei acts on declared relations, so we measured whether agents can propose them. We used citations in mathematics papers from a pinned arXiv snapshot. A unit is one citing work and one cited result; its label states whether the citing work depends on the cited result.

In stage A, two reviewer models from different vendors labelled 160 units independently from the citing passage. They agreed on 143 (Cohen's kappa 0.75 over all labels). A citation that names a specific numbered result is a weak signal: of 48 agreed units with such a locator, 20 depend and 28 do not.

Stage B selected the 136 units whose complete source was at most 160,000 bytes, from 92 citing works. It excluded 24 of the 160 stage A units under that source size rule. The exact source bytes are pinned, but their arXiv version labels remain unproven. The experiment provides the citing source, not a verification of the cited theorem. The two reviewers labelled these units again from the whole paper and agreed on 118, of which 60 depend. Both methods in the table used the same proposal model, google/gemini-3.8-flash; the reviewers were anthropic/claude-sonnet-5.5 and openai/gpt-6-sol.

Table 7: Dependency proposals on 118 agreed mathematics units (60 dependencies).

Method	Dependencies found	False alarms	F1
Reads the citing passage, 20 units per request	32 of 60	1	0.688
Reads the whole citing paper, one work per request	44 of 60	0	0.846

The paired F1 gain is 0.158, with a 95% interval of 0.067 to 0.256 (2,000 resamples by citing work). Scored against each reviewer alone, the gain is 0.170 and 0.119, and neither interval includes zero. The whole-paper method cost USD 2.57 for 92 model calls. This is the proposal model's cost alone. The two reviewer models cost USD 15.61 for 184 further calls. The three roles total USD 18.18 for 276 completed calls in the retry run. These provider totals exclude the earlier stage A calls, collection, human review, and infrastructure.

This comparison does not isolate the effect of reading more text: the two methods also differ in prompt and in batching. The reference labels are agreement between two models that read the whole paper, not expert judgment, so the passage method is scored against labels made from text it never saw. With full text, the reviewers moved 21 units from "does not depend" to "depends" (a post-hoc count). The units come from the same inspected population as the calibration, with no held-out set, and we did not test dependency proposals on code. Any system, including the equipped reference, can call the same model, so this result concerns dependency capture, not Kyoei. The whole-paper method still missed 16 of the 60 agreed dependencies. A person must therefore confirm each proposed relation against the passage that supports it. A node also cannot treat "no relation found" as independence: work with an undeclared dependency receives no notice when its source changes.

5.7 Withdrawal across separate nodes

Three institution processes each keep their own SQLite store. They declare 136 claims from the stage B units with the whole-paper labels, 36 derived claims and 6 joint sections, then process challenges, a withdrawal, dissent and repairs. Every challenge and the withdrawal are simulated, and the derived claims and joint sections are synthetic links, so none establishes a failure in a cited paper. The withdrawal is of an outside source, and the fixture accepts it from any trusted institution. The run therefore tests neither owner-only withdrawal nor independent confirmation of an external retraction. Events cross between processes only as signed HTTP requests on the loopback interface, and each process computes its own affected work from its own records. The node clocks are fixed during the run, so event expiry is not exercised.

All 869 requests were signed, and none was refused: 15 setup requests, 332 declaration requests, 12 repair requests, 137 external events and 373 observations. 432 signed events crossed between processes on their first hop, none refused and none undeliverable. Against the frozen expected outcomes, the scorer found no missed delivery, false alarm, unexpected record or continuity break on the agreed-label subset. All 31 expected derived claim/event pairs had a notice. In a separate scripted repair sequence, one cited result was challenged and three institutions revised the work that relied on it; its 3 notices matched the expected values on all six fields. Sixteen dependencies that the labels missed produced no notice, because nobody declared them. A rule that flags every citing claim would have raised 58 false alarms on claims that do not depend on their source.

The scorer counts only the 118 units on which the reviewers agreed for accuracy. Of 18 disputed units, one received a notice and 17 remained silent; silence on these units is not evidence of safety. The leak check covers unexpected records, notice recipients and inbox subjects. It does not test traffic analysis, prompt disclosure or every query a hostile peer could make.

The batch scorer checks claim/event pairs but not each notice's version. It also misses duplicate records and some notices assigned to the wrong challenge: an in-memory test that added each kind left its error counts at zero. A separate paper check compares all observed batch pairs with the declared expected set. The saved batch has 76 records for 75 distinct claim/event pairs, with no repeated record for the same version; one derived claim appears at two versions. This recount finds no extra pair, but it does not validate each version against an independent repair oracle.

This is mechanism evidence, not a comparison, because the rebuilt reference arm did not pass review. The run stays marked as in repair until a separate review accepts its driver. Two prototype gaps also limit it. The outbox never recovers a claimed entry at startup, so a crash between claiming and sending

would leave the event unsent; this run had no crash. The store can also accept new reliance without consulting its block table, so a reported blocked state does not establish refusal at every use interface.

5.8 The Cloudflare port

The port is TypeScript code that we wrote for the contest. The following implementation description comes from the reviewed source snapshot in `alignment.json`; the saved deployed runs have no build digest. We cannot bind that snapshot to the historical deployment. Agent runs execute inside the organization's *OrgNode* Durable Object, with no separate object per run, so an interrupted run does not resume by itself. An alarm retries queued events with exponential back-off, capped at 30 seconds. A run's write token lives one hour. Releases last at most 24 hours, and an onward release cannot outlive the grants of its inputs. Artifacts tokens last at least 60 seconds, so a read token can outlive a release that has less than a minute left. Agents use a Workers AI model with output bound to a JSON schema, and a second model when the first answer fails; the node records the first model's name for both. Signatures use ML-DSA-65 through WebCrypto. Workers expose ML-DSA only behind the opt-in `webcrypto_modern_algorithms` compatibility flag, and the interface may still change.[35] Each organization has a command-line client and a terminal view (`watch`). Together they show what changed, what is affected, why, and what can be done next; the client's commands include `release`, `approve`, `gate`, `ship`, `withdraw`, `challenge`, `alternatives` and `rebase`.

The port records a claim through its task, rationale and committed files; it has no separate claim or evidence-validation field. The scenario supplies the dependency relations and interface labels. An agent reads released inputs, generates and commits its edit, then registers reliance with their owners. Registration does not prove that the generated edit implements those relations.

The current port passes 46 local tests, including the original acceptance suite and the release regression tests. The tests run three nodes with separate stores and keys, an in-memory transport, and a local Git service. They sign with Ed25519 and establish local behavior, not deployed fault recovery.

The release review found failure paths that the original 25 tests did not cover. The current source fixes them as follows:

- The node stores a state change, its inbox receipt, and the delivery work it causes in one transaction. The dispatcher signs the saved work afterwards. Tests restore a saved store at the former crash windows and check that the resulting events still arrive. Failed token revocations also survive restart and retry until confirmed or until the token lifetime ends. The Worker arms its alarm before work starts and keeps a 30-second alarm after first use, including idle periods. That idle alarm wakes the Durable Object every 30 seconds even when no work is queued. No test has crashed a deployed Durable Object to check this recovery.
- The ship gate checks grants for the declared builds-on closure, including expiry, purpose, and verified snapshots. It refuses a foreign candidate or input whose reliance registration is pending. Tests cover an expired transitive input and an unavailable candidate owner.
- An owner answers reliance only for a recipient covered by its own grant and purpose. A former recipient can still receive the withdrawal state. Dissent goes to the notice's source, instead of every pinned peer. Tests check both the permitted exchange and refusal of an unrelated peer.
- The port's Artifacts adapter writes the operation note before it updates main. A failed note write therefore cannot leave an unmarked merge. A failed main update can leave an unused note. Reconciliation remains unresolved if another change has moved main beyond the operation under review. The local Git service of the tests and the offline demonstration still updates main before it writes the note.
- Forwarded challenge notices retain the withheld-evidence marker. Peer addresses must use HTTPS, except for the local loopback demonstration. Administrator requests create task delegations for at most one hour. The node rechecks the person, task, declared inputs, file list, purpose, expiry, and single-use limit before registering an agent's contribution.

Additional tests cover concurrent reliance, an older valid reply, duplicate causes, symlink targets, submodules, and unsafe paths. Some original tests still assert less than their names state; Table 3 marks them as weak. The port does not create isolated export repositories for forks with undisclosed history. Repository read tokens can expose that history, so such repositories must not be released. People still act through administrator tokens rather than personal signing keys. HTTP routes, model calls, and Worker alarms need deployed fault tests. These source fixes do not change the historical run logs or show that the historical deployments contained the fixes.

The port's repository runs the scenario without a Cloudflare account. It needs Node.js 22.18 or later and `git`. The commands `npm ci`, `npm test` and `npm run demo` install the pinned packages, run the tests and run the full scenario on three local node processes. Two commands deploy the three Workers and run the scenario on them: `node scripts/deploy.ts up <env> --subdomain <subdomain> --model workers-ai`, then `npm run demo -- --remote <env>`. The scenario script uses the client's own commands; Acme's repair onto Cove's alternative, for example, was `kyoei rebase a3 --org a --onto c/c5 --agent a3r`.

5.9 Deployed runs on Cloudflare

We deployed the port as three Workers, one per organization, in one Cloudflare account. We first ran the scenario with scripted edits (`deployed-1`), then with Workers AI agents (`deployed-2-ai`). A *line* is the sequence of versions of one contribution.

Figure 2 shows the experiment's deployment and request paths. The scenario script controlled all three organizations; they did not operate independent accounts.

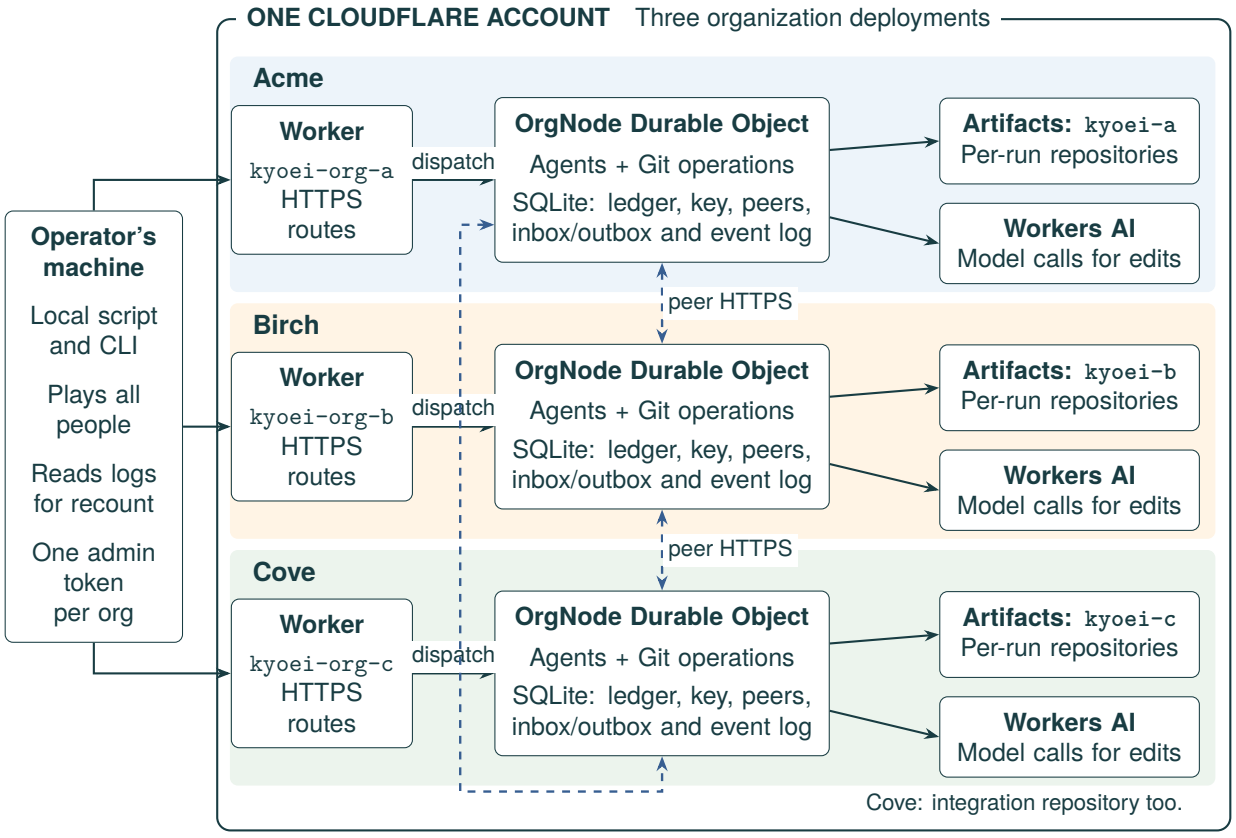


Figure 2: Cloudflare experiment architecture. Each Worker dispatches to one OrgNode, which runs agents and uses its own state, Artifacts namespace and AI binding. Peer exchanges use public HTTPS. The local script acts for every person and collects the logs.

Table 8: Architecture of the deployed experiment. Parts in the first four rows exist once per organization.

Part	Where it runs	Role in the runs
Worker `kyoei-org-a`, `-b`, `-c`	Workers, own `workers.dev` URL	Dispatches application requests to one OrgNode; `/health` responds at the Worker
OrgNode Durable Object	Durable Objects with SQLite storage	Holds the ledger, the signing key, pinned peers, outbox and event log; runs the agents
Artifacts namespace `kyoei-a`, `-b`, `-c`	Artifacts	One repository per agent run; Cove's namespace also holds the integration repository
Workers AI binding	Workers AI	Writes each agent's edit as JSON bound to a schema
Scenario script	Operator's machine, Node.js	Plays every person through the administrator routes, then fetches the three logs

The routes separate peers from people. Peers call three routes over public HTTPS: `/kyoei/identity` returns the node's public key, `/kyoei/inbox` accepts signed events, and `/kyoei/rely` registers a reliance. People call `/api/` routes with a bearer administrator token; each Worker stores only the SHA-256 digest of its token as a secret.

In each run, the script first fetches each node's identity over HTTPS and pins the other two fingerprints at each node. The run therefore did not exercise an exchange out of band. The script starts agent runs through the administrator API, with one request per organization for each wave and separate requests for two later runs and each repair. The OrgNode runs each agent in the background as section 4.2 describes, and Git operations use isomorphic-git on an in-memory file system inside the Durable Object. At the end, the script clones the integration repository with plain `git`, fetches its notes, then reads each node's log through `/api/log`. Throughout, the script played every person: it released, approved, shipped, withdrew and chose repairs, so no person reviewed a change. The script did not restart an agent run, take a peer offline or let a release expire.

The scenario script supplied each task, dependency relation, and alternative interface label. Agents generated edits; they did not discover dependencies or choose the repairs. Six first-wave run lifetimes overlapped in the logs. Cove's node flagged two Acme intents on the same file, src/retry.ts, before either agent finished; no run or test flagged an overlap between agents of two organizations. The first of the two changes shipped. The gate refused the second with `stale-base` and the reason "main changed src/retry.ts after this base". A repair agent rebased it, the script approved it again as Cove's person, and it shipped.

Birch released its signer to Acme and Cove and allowed Acme to pass work built on it to Cove; Cove released an alternative signer to Acme. In the second wave, one line of work at each organization built on Birch's signer, and one Birch line cited it. The script then simulated a licence problem and withdrew Birch's signer. No model or licence checker detected this problem. The *incident* is the interval from that withdrawal to Birch's replacement release.

At the withdrawal, Birch's node revoked the two read tokens that it had issued for the signer and sent a signed event to each peer. Each node then computed its own affected work. Revoking those tokens did not remove read access to the signer's first version, which had already shipped to the integration repository; agents read that repository through a separate release token. The nodes blocked managed reuse of the two shipped items instead: Birch's signer and Cove's line built on it. During the incident, Cove's gate refused Acme's dependent line with `withdrawn` and Birch's reason, "origin of one snippet is not verified; license unknown". An unrelated Cove line started just after the withdrawal and shipped during the incident.

A repair agent moved Acme's blocked line onto Cove's alternative. Birch then released a newly generated replacement, and two more repair agents moved the other blocked lines onto it. No step checked the replacement's licence, and no build or test ran on any repaired line.

Table 9: Recorded Cloudflare run `deployed-2-ai`. The values come from the three node logs. `verify.py` recounts each value from those logs, separately from the summary script of the port.

Quantity	Value
Agent runs; failed	16; 0
Runs labelled with model `@cf/moonshotai/kimi-k2.7-code`	16 of 16
Overlapping logged agent-run lifetimes, maximum	6
Same-file overlaps flagged before either agent finished	1 of 1
Releases received with the BLAKE3 root recomputed from the fetched commit	14 of 14
Queued signed events delivered between nodes; refused; retried	50; 0; 0
Reliance registrations with an owner reply, outside the queued count	13

Quantity	Value
Read tokens issued in releases; revoked at the withdrawal	14; 2
Maximum withdrawal queue-to-receipt time, owner's clock	65 ms, first attempt
Lines of work blocked by the withdrawal at Acme; Birch; Cove	1; 1; 1
Shipped items blocked in the integration repository	2
Citing lines with a notice and no block; shipped during the incident	1; 1
Unrelated lines shipped during the incident	1
Ship-gate refusals during the incident	1
Repairs onto another organization's alternative; onto the owner's replacement	1; 2
Withdrawal to the merge of the repair onto the alternative	143.2 s
Replacement release to the last repaired merge	93.8 s
Merges refused with `stale-base`, then repaired	1
Person actions: approvals; releases; withdrawals	15; 11; 1
Notices open at the end, all without a block	2
Run duration	589 s

The 50 deliveries count queued events only. Both runs also record 13 synchronous reliance request/reply exchanges, outside that count. Neither count includes administrator requests, Git transfers or model calls.

A peer answers "applied" after checking the event and opening its notices. The 65 ms starts when the sender creates each outbox entry and ends at its receipt. It excludes earlier owner processing, token revocation, and envelope signing; it is not withdrawal-command latency. This interval uses one clock. Overall elapsed time, peak overlap, and repair times combine different nodes' wall clocks without measuring their offsets. Peak overlap counts preparation, model calls, and Git work, not simultaneous model inference.

Each agent run took 15.5 to 115.4 s. The repair times include model latency and the script's pacing, not human response time. Both open notices belong to the citing line; a nonblocking notice has no closing rule yet. The run with scripted edits gave the same counts for every row except the edit source and the times.

These logs are node reports, not cryptographic transcripts: they omit the signed envelopes needed to reverify delivery signatures offline. The saved identities and matching pinned fingerprints do not replace those envelopes. For both runs, the checker matches each queued event to one sender receipt and one receiver entry by event, issuer, recipient, and kind. This checks log consistency, not authenticity. The evidence also lacks a deployed build digest, so current source inspection cannot establish which exact Worker bundle produced the logs.

The run exercises coordination on Workers, Durable Objects, Artifacts, and Workers AI. It has no comparator, independent accounts, reviewed code, or failure injection. Model outputs are not deterministic, and each value is one observation. Separate-account operation still needs routing, trust, access, and recovery tests.

6 Conclusion and perspectives

In this section, we state what the results establish and which obstacle remains. We then give the perspectives we plan to pursue.

6.1 What the results establish

In simulated collaborations that we designed, the research prototype supplied the collaboration controls without code written by the adopting team, and it matched our equipped reference on every fault trial. Supplier work in the second collaboration and higher setup costs prevent a claim of lower operating effort. The dependency experiment measures model proposals against model agreement; it establishes neither claim correctness nor complete dependency capture.

In one Cloudflare account, the deployed run took concurrent agent edits, scoped releases and one withdrawal through signed records and gate checks. Repair agents then moved the blocked work onto an alternative or onto the owner's replacement. Anyone can rerun its scenario from the companion repository, although model outputs will differ. The run does not establish reliable recovery, autonomous repair selection, or usability for independently operated organizations. The immediate work is to close the remaining port gaps, bind a new run to a deployed build, and measure the effort of real teams.

6.2 Perspectives

Installable nodes and pilots. The deployed run used one account, one operator and administrator tokens. Independent teams need separate accounts, personal keys with rotation, agreed peer pinning, and confirmation of proposed dependencies. Pilots must then measure setup, repair time, support effort, quality, and willingness to pay.

Behaviour checks at the gate. The gate checks records, terms and bases, not the behaviour of the code. A later gate can require the integrator's own build and test results, bound to the approved commit, before a merge.

Verifiable run records. The saved logs omit the signed envelopes and the deployed build digest. A later run will store both beside its logs, so that a reader can check delivery signatures and the executed code offline.

One view across collaborations. An organization in several collaborations needs one answer to "which of our work depends on this change?". Separate consortium systems can answer it through a shared query or a bridge, so a fair comparison must let the reference reuse such an adapter.

Research repair at scale. The dependency experiment compared model proposals with model agreement on mathematics papers. We will extend full-text proposals to dependencies that experts review, and evaluate repair across independently operated research groups.

Results without disclosure. A proof that a private table satisfies a stated property would let another organization decide without seeing the table. This extension depends on a hash-based proof system that is still under development.

Native BLAKE3 Git, standards and mechanized properties. A native BLAKE3 object format must preserve bytes and topology and state which signatures a conversion invalidates; no current result shows a speed or security gain over SHA-256. We also plan to publish the record and envelope formats, mapped to PROV and the Citation Typing Ontology, and to check the affected-set rule in a proof assistant.

Use of AI tools

AI coding agents wrote most of the code and drafted parts of this paper under the direction of the author, and AI reviewers red-teamed the draft. Section 5.2 states what the paper's checker, `verify.py`, checks and what it does not.

Conflict of interest

Slyseed, Inc. and Lemma Ventures AG maintain Kyoei. This paper accompanies an entry to the Cloudflare competition "Build the Next-Gen Git Platform on Cloudflare".

Data availability

The companion repository, `sinequanone-org/kyoei`, contains the TypeScript port, local tests, demonstration, paper source, and this PDF under Apache 2.0. Its `evidence/` directory contains the saved reports, frozen contracts, and Cloudflare run logs used by the paper. `paper/verify.py` checks the reported values and file bindings. The repository also identifies archived source snapshots where a historical report binds a different version from the current source.

The selected Python files support evidence checks; they are not a runnable release of the research prototype. Third-party source passages, raw challenge records, model receipts, credentials, and private research records are excluded. The dependency-proposal experiments cannot be reproduced from this bundle alone. The saved Cloudflare logs contain neither signed envelopes nor a deployed-build digest; they support consistency checks, not independent authentication of the historical deployment.

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